

Major Project

Proposal Form

Complete this form before your proposal presentation and have it to hand during the session. Upload the completed form to Learning Space alongside your presentation slides. Your supervisor will add their feedback at or following your first supervision session after the presentation. The form is a record of the key commitments made in your proposal; it is not a substitute for the presentation itself.

1. Student details

Full name	██████████
Student number	██████
Programme	MA Game Design
Supervisor	██████████
Date of proposal presentation	24/06/25

2. Project title and research question

Proposed project title	Multiple-Solution Puzzle Design
Research question	1. How does the puzzle design process differ for multiple-solution puzzles from single-solution puzzles? 2. What are the unique affordances of multiple-solution puzzles when compared to single-solution puzzles?

3. Artefact

Artefact type	Digital game
Artefact description and intended audience	The levels I will create for the digital game: Window Adventure Game. It is a single-player desktop cosy

puzzle adventure game where players manipulate the operating system user interface to guide a character through a fragmented world. The primary objective is to connect a series of illuminated checkpoints scattered across separate windows. Successfully aligning these windows creates an uninterrupted route, opening the exit portal for the character.

4. Field context and key references

Field context

The contemporary state-of-the-art in digital puzzle design is heavily anchored in linear, single-solution paradigms. These paradigms rely on a carefully calibrated loop of tension and release. Foundational frameworks establish that the primary driver of player satisfaction is a “Eureka” moment which is the psychological flash of insight and joy experienced upon discovering a solution. For this moment to resonate, designers must guide players through a structured cycle of cognitive resistance. Selinker and Snyder (2013) describe the process as the mastery of frustration, where the ultimate reward stems directly from being “released from the pain” of a highly frustrating puzzle. To maintain this delicate balance without inducing outright alienation, designers leverage formal scaffolding frameworks. Schell’s (2020) 10 design principles highlight the necessity of cultivating player engagement and trust; the player must remain confident that the puzzle is fair and solvable through logic rather than trial-and-error. Mechanically, this trust is enforced via checkpointing; the architectural practice of breaking a complex macro-puzzle into a series of identifiable, self-confirming micro-steps. Such designs are traditionally executed through approaches such as:

- Forward Design (Grant, 2021; Grant, 2025), which sequentially introduces specific pieces of information exactly when the player requires them to deduce the next logical step.
- Narrative-First Design (Barth & Messinger-Michaels, 2019), which nests mechanic limitations directly inside the thematic or storytelling boundaries of the game world.

However, the materials focus on the notion of a single solution. A significant gap exists in academic and industry literature regarding approaches for designing multi-solution puzzles.

Key references

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- Schon, D.A (2017) The Reflective Practitioner: How Professionals Think in Action. New York: Routledge.
- Selinker, M. & Snyder, T. (2013). Puzzle Craft: The Ultimate Guide On How To Construct Every Kind of Puzzle. New York: Puzzlewright Press.
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5. Methodology

Research and development approach

This practice-based research project is situated within a research-through-design framework (Zimmerman et al, 2007), investigating player engagement by comparing linear versus multi-solution puzzle design in a novel, window-overlapping spatial mechanic game. This process aims to deliver a functional prototype and converge upon a formal design framework for developing expressive, multi-solution puzzles. Beyond practice-as-research (see Kaszynska & Kimbell, 2025), I propose a within-subjects A/B testing model (Quin et al, 2024) to investigate the psychological and cognitive impacts of epistemic agency. By observing human interactions within an open-ended systemic environment, I seek to understand how players navigate failure dynamically, exercise creative problem-solving autonomy, and find meaning in emergent outcomes. Because these subjective experiences of cognitive ownership, frustration, and aesthetic satisfaction cannot be inferred from the designs themselves, involving people is vital to capturing and exploring how design choices influence human psychological resilience and learning behaviours. Though, reflection in action is important to refine the framework (Schon, 2017; Nelson, 2022).

6. Ethics

Full ethics application required?

Yes – Medium risk

Ethics considerations

To comply with the university ethics policy and procedure, I consider legal, social, ethical, environmental, and professional standards as part of my project proposal, particularly acknowledging that playtesting involves human participation and is therefore classified as a medium-risk activity. Legally, the project ensures full compliance with GDPR (including relevant cybersecurity protocols including encryption and password protection) in processing personal data, intellectual property laws (particularly, copyright and

design right), third-party service Terms of Service (e.g., Discord), and the Health and Safety at Work Act through rigorous risk assessments (e.g., display-screen use and eye strain/posture, computer interface use and repetitive strain injury, etc.). Socially and ethically, the research prioritises digital accessibility (A11Y) and neurodiversity accommodations alongside community safeguarding, while securing informed consent, managing deception or framing, and ensuring I only recruit (Gillick) competent participants (typically, age 18+). From a sustainability perspective, the study acknowledges its digital carbon footprint, sets relatively low hardware access requirements, and makes little to no use of excessive energy consuming tools or processes. Professionally, the work is guided by the IGDA's Core Values and Code of Ethics and the DiGRA Code of Conduct, academic integrity via the Concordat to Support Research Integrity, and I declare no conflict of interest. I also strive to limit the scope of any agreement and non-disclosure requirement associated with the project to only those necessary and permitted under the 2018 Trade Secrets Regulations. Because playtesting directly relies on human subjects, a steadfast duty (i.e., the "neighbour principle") is maintained throughout the process to mitigate the inherent risks to protect the rights and safety of all participants.

7. Project plan

Complete the table below with your key milestones and any anticipated blockers. You are not expected to fill every row — use as many as are meaningful for your project.

Week	Key milestone or deliverable	Potential blocker / contingency
Week 5	Building the foundational infrastructure and implementing the core mechanics	Tech challenges – seek support from colleagues in dev team
Week 7	Designing the first set of puzzles	Creative blindspots – agile approach, sharing with peers as I progress
Week 9	Play-testing and reflecting on puzzle design methodology	Ethics / Recruitment – seek advice from supervisor
Week 11	Designing the second set of puzzles	Creative blindspots – agile approach, sharing with peers as I progress
Week 13	Play-testing and reflecting on puzzle design methodology	Ethics / Recruitment – seek advice from supervisor
Week 15	Presenting findings	Network connection – plan ahead to ensure a good place to present from

8. Process documentation

You are required to document your development process throughout the module. The method you choose should be appropriate to your artefact type and accessible to your assessor as part of your final submission. Your supervisor will confirm whether your proposed method is appropriate.

Proposed method	One-page design document, versioned repository, reflective report
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Justification

Following Nelson's (2022) model of multi-modal practice research, I propose a three-part structure including a one-page design document, the digital artefact itself and its documented previous versions, and a short report. I believe this suitable for a master's level project because it mirrors professional interactive media workflows while maintaining rigorous academic validity via research-through-design. The one-page design document provides a concise, industry-standard synthesis of the design approach, offering a transferable design framework for the wider game development community. Simultaneously, the versioned repository serves as the empirical core of the project, with the commit history providing an audit trail of level design iterations to make the research and development open and transparent. The reflective report contextualises these practical outputs within established design literature, offering a critical space to reflect upon the play-testing data and articulate how the act of making the levels generated new, systematic knowledge regarding a method for multi-solution puzzle design.

Supervisor feedback

Please provide feedback on each section of the proposal, including specific comment on the proposed process documentation method. This form should be completed at or following the first supervision session after the proposal presentation.

Supervisor signature / confirmation:**Date:**